Chemical Engineering Journal 489 (2024) 151439

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

The multiple enhancements of deep eutectic solvent on cellulase significantly improve the saccharification efficiency of lignocellulosic biomass

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ARTICLE INFO

Keywords:
Cellulase
Deep eutectic solvent
Enzymatic hydrolysis
DFT calculation
MD simulation

ABSTRACT

The introduction of specific choline chloride (ChCl)- and betaine (Bet)-based deep eutectic solvents (DESs) into the conventional acetic acid–sodium acetate solution led to a significant increase in cellulase activity and stability. In particular, a higher enhancement in β-glucosidase activity was obtained, which plays a key role in improving cellulase function. The DESs prolonged the half-life of cellulase. The addition of 10 % (v/v) ChCl/1,3-propanediol (1:2) and 5 % (v/v) Bet/glycerol (1:2) resulted in the most significant improvement, with the half-life of cellulase reaching three times the initial half-life value. The above both DESs allowed the cellulase to maintain an excellent level of activity over time, even after 72 h. The DESs tightened the structure of the enzyme by changing its secondary structure, which had a stabilizing effect on the overall conformation. The density functional theory calculations of DESs revealed that the DES with a more stable structure more effectively protected the active conformation. The molecular dynamics simulation on β-glucosidase indicated that the hydrogen bond donor end of the DES primarily formed hydrogen bonds with the enzyme, which induced a spatially compact conformation by increasing the α-helix content while reducing the β-sheet content. It also restricted the movement of amino acids, particularly stabilizing the active site, and increased the hydrophilic region, thereby enhancing enzyme activity and stability. In addition, the DES formed a protective layer that prevented the unproductive adsorption of lignin on the enzymes during the hydrolysis of lignocellulosic feedstocks. These multiple improvements provided by DESs significantly increased the enzymatic saccharification efficiency of corn stover and poplar.

1. Introduction

Lignocellulosic biomass, the most abundant biomass resource on Earth, is green, clean, and renewable; thus, it holds great potential for addressing the resource scarcity and energy crisis facing human society [1,2]. Efficient conversion of cellulose-rich substrates to fermentable glucose through the sugar platform method is an important pathway for resource conversion [3]. The sugar platform conversion pathway involves pretreatment to increase the accessible surface area of the substrate and separate cellulose, and conversion of the separated cellulose to fermentable glucose for biofuel production [2,4]. As is well known, the enzymatic hydrolysis process using cellulase is the critical step [5].

The traditional cellulase-catalyzed hydrolysis system operates within an acidic buffer solution at pH 4.8. However, cellulase is prone to inactivation and displays limited stability in this reaction environment [6]. The instability and low activity of cellulase pose a major challenge to the application of hydrolysis technology on an industrial scale [7,15]. Ionic liquids are considered as green alternatives to traditional solvents and have been extensively studied in biotransformation processes [8]. However, the high price, lack of biodegradability, poor sustainability,

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https://doi.org/10.1016/j.cej.2024.151439
Received 19 February 2024; Received in revised form 8 April 2024; Accepted 17 April 2024
Available online 18 April 2024
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complex preparation and high cost of ionic liquids limit their use [9]. Deep eutectic solvents (DESs) are a new type of green solvents developed after ionic liquids [10,11]. They are formed by mixing hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs) in precise proportions and then heating to generate a new liquid with a hydrogen bond network structure [12,13]. DESs are characterized by good biocompatibility and a stable enzyme molecular structure, which helps to solve the application bottleneck caused by environmental mismatch in the enzymatic catalysis industry, providing a suitable medium environment for enzyme-catalyzed reactions [12,14–17].

Studies have shown that the strong hydrogen-bonding network structure of DESs can stabilize the conformation of enzymes, thereby enhancing their catalytic activity and stability [18–20]. Common enzyme systems such as lipases, proteases, and hydrolases have demonstrated improved activity in DESs. Yu et al. [21] found that pancreatic trypsin rapidly bound to substrates and improved the reaction efficiency in a 50 % DES system composed of sucrose and betaine (Bet) at a 1:7 ratio. The presence of a hydrogen bonding network can stabilize the active sites of enzymes in different environments. Khodaverdiana et al. [22] found that the activity of laccase in Bet/ glycerol (Gly) (1:2) was higher than that in single-component aqueous solutions, because the DES featured a strong hydrogen-bonding network structure that could maintain the integrity of the laccase structure. Esquembre et al. [23] found that the thermal unfolding of lysozyme in DES–water dilutions was reversible, and the thermal effect on lysozyme in DES–water dilutions was minimal, indicating that the DES had a protective effect on the lysozyme structure. Ibrahim et al. [16] found that the introduction of 40 % (v/v) choline chloride (ChCl)/ethylene glycol (EG) into a phosphate buffer improved lipase activity by 230 % compared with the pure buffer solution, but lipase was inactivated in the pure DES system. Astrid et al. [24] found that the activity of laccase in water solutions containing 25 wt% of a DES was almost 10 times that in the DES-free water solution. These findings further demonstrate the benefits of DESs in improving the activity and stability of enzymes.

Cellulase, a key catalyst in cellulose hydrolysis, plays a crucial role in biomass conversion. An efficient cellulase-catalyzed hydrolysis system can effectively convert cellulose into monosaccharides, which can then be further converted into biofuels or chemicals. However, the industrial application of cellulase is often limited by insufficient stability and activity, hindering commercialization [24]. The application of DESs has been considered a promising strategy to improve cellulase performance owing to the unique ability of the solvents to regulate enzyme stability and activity. Few studies have shown that DESs can improve cellulase performance. Gunny et al. [25,26] found that cellulase maintained over 90 % activity in ChCl/Gly and ChCl/EG hydrolysis systems. Xu et al. [27] found that β-glucosidase activity was significantly enhanced in a novel DES composed of ChCl/1,2 propylene glycol and 6 % (v/v) water compared with the activity in a buffer solution. Yao et al. [28] investigated the effects of Bet and ChCl mixed with various organic small molecule compositions of DESs at different concentrations on cellulase activity and found that diols exhibited superior activity. However, research on the application of DESs to improve cellulase stability and activity is relatively scarce. Further investigations are needed to elucidate the mechanism through which DESs affect the activities of different components of cellulase, the role of DESs in the catalytic process, and the performance of DESs in practical biomass feedstocks. These in-depth studies will enable the development of more suitable DESs for industrial applications, which will effectively promote the advancement of biomass conversion technology, the production of renewable resources and clean energy, and the sustainable development of the bio-economy.

ChCl and Bet contain functional groups capable of accepting protons and are commonly used as HBAs for DES preparation [12,29]. Moreover, they are inexpensive, widely available, and easy to prepare, making them suitable for industrial use. In addition, these HBAs are widely present in biological systems; for example, ChCl is used in animal feed, and Bet is a natural product extracted from sugar beet, making them nontoxic and readily biodegradable [30,31]. Given these properties, ChCl- and Bet-based DESs have great potential for application in the field of biocatalysts. In this study, different types of DESs were synthesized using ChCl and Bet as HBDs, and acids, bases, and alcohols as HBAs. The DESs were introduced into a conventional acetic acid–sodium acetate (AA–SA) solution to construct a novel DES/AA-SA hydrolysis system for cellulase. The effects of the DESs on the catalytic activities of cellulase and its main components and on the stability of cellulase were investigated. Fluorescence spectroscopy and circular dichroism were employed to detect conformational changes in enzymes. Kamlet–Taft values and density functional theory (DFT) calculations were used to study the hydrogen bonding type and structural properties of DESs. Molecular dynamics (MD) simulation was employed to analyze the structure–performance relationship at the molecular level, to elucidate the mechanism through which DESs enhance the activity of cellulase. Additionally, the unproductive adsorption capacities of lignin on cellulase in the DES/AA-SA solutions were investigated.

The results indicated that DESs offered several enhancements through hydrogen bonding interactions with cellulase. 1) They increased activity by inducing a secondary structure change, stabilizing the active site, and expanding the hydrophilic region. 2) They enhanced stability by compacting the spatial structure, thereby reducing molecular vibration and friction and slowing conformational degradation over time. 3) They formed a protective layer on the enzyme surface, which reduced the unproductive adsorption of lignin on enzymes. These combined enhancements significantly improved the saccharification efficiency of lignocellulosic biomass, increasing glucose yields from 78 % to 98 % for certain substrates. Therefore, this novel hydrolysis system has considerable potential for promoting the development and application of biomass conversion technology in industrial platforms.

2. Materials and methods

2.1. Materials

Cellulase and β-glucosidase (βG) were purchased from Novozymes (China) Biotechnology Co., Ltd. The pulping enzyme was provided by Asia Pacific Senbo (Shandong) Pulp and Paper Co., Ltd. ChCl and Bet were purchased from Shanghai McLan Biochemical Technology Co., Ltd. 4-nitrophenyl-β-D-glucopyranoside (pNPG), and 4-nitrophenyl-β-D-cellobioside (pNPC) were purchased from Aladdin. while the remaining reagents, including glycerol (Gly), 1,3-propanediol (1,3-prop), 1,2 propylene glycol (1,2-prop), ethylene glycol (EG), polyethylene glycol 200 (PEG200), urea (U), lactic acid (LA), carboxymethyl cellulose (CMC), acetic acid (AA), sodium acetate (SA), 1,4 butanediol (1,4-bu), were obtained from China National Pharmaceutical Group Chemical Reagent Co., Ltd.

2.2. The preparation of DES

ChCl and Bet as HBAs were mixed with different types of HBDs, including acids, bases, and alcohols, at specific ratios (1:1, 1:2, 1:3, 1:4, and 1:5), and the synthesis reaction was conducted at 80℃ and 400 rpm until a clear and transparent liquid was formed as the DES [13]. After the DESs were left to stand at room temperature for 1 h, the stable solutions were selected for further processing.

2.3. The enzyme activity

The synthesized DES was mixed with an AA–SA solution (pH 4.8, 0.05 M) to form a 10 % (v/v) DES/AA-SA solution to study the effects of different types of DES on cellulase activity.

Cellulase activity was measured using the method published by the National Renewable Energy Laboratory (NREL) [32]. The total cellulase activity was represented in terms of filter paper activity. The specific procedure was as follows: a Whatman No. 1 filter paper strip (1.0 × 6.0

cm) was soaked in a 1.0 mL solution of DES/AA-SA containing a 0.5 mL enzyme appropriately diluted in the same solution. After rapid agitation, the reaction was conducted at 50 °C for 1 h. Then, 3.0 mL of 3,5-Dinitrosalicylic acid (DNS) solution was added to halt the reaction, and the mixture was boiled for 5 min. The absorbance of the solution in the tube was measured at a wavelength of 540 nm, and the content of reducing sugars was determined using a linear regression equation. This procedure is designed to measure cellulase activity in terms of filter paper units (FPU) per milliliter of original (undiluted) enzyme solution.

$$FPU = \frac{0.37}{\text{Enzyme concentration to release 2.0 mg glucose}} \text{ units mL}^{-1} \quad (2.1)$$

Endo-glucanase (EG) activity was measured according to the IUPAC method [33]. A suitable enzyme (0.5 mL) and 1 % carboxymethyl cellulose (CMC) (1.5 mL) were dissolved in citrate buffer (pH 4.8), and the solution was added to a test tube with a volume of at least 25 mL, mixed well, and incubated at 50 °C for 30 min. At the end of the incubation, 1.5 mL of DNS was added to the solution, which was then boiled for 5.0 min in a vigorously boiling water bath. Afterward, the test tube was transferred to a cold-water bath. The volume in the tube was adjusted to 25 mL with deionized water, and the absorbance intensity was measured at 540 nm. One unit of EG activity (1 U/mL) was defined as 1 μmol glucose produced in 1 min under hydrolysis conditions of 50 °C and pH 4.8 for 30 min with the CMC substrate.

Exo-glucanase/cellobiohydrolase (CBH) activity was determined using pNPC as the substrate [33]. A 1 mg/mL solution of pNPC (1.8 mL) was dissolved in citrate buffer (pH 4.8), and a suitably diluted enzyme (0.2 mL) was added to the solution. The obtained solution was then incubated at 50 °C for 30 min. Afterward, 2 mL of 2 % Na_2CO_3 was added to terminate the hydrolysis reaction. The hydrolysis product, *p*-nitrophenol, was measured at 400 nm. One unit of CBH activity (1 U/mL) was defined as the amount of enzyme liberating 1 μmol of *p*-nitrophenol per minute under the assay conditions.

The βG activity was determined using pNPG as the substrate [33]. A 5 mM pNPG solution (1 mL) was dissolved in citrate buffer (pH 4.8), and 0.5 mL of a suitable enzyme was dissolved in the solution. The obtained solution was then incubated at 50 °C for 10 min. The hydrolysis reaction was terminated through the addition of 2 mL of 1 M Na_2CO_3 solution, and then 10 mL of deionized water was added to the solution after cooling. The resulting *p*-nitrophenol was measured at 400 nm. One unit of βG activity (1 U/mL) was defined as the number of μmol *p*-nitrophenol produced per minute under the assay conditions.

2.4. The stability of cellulase

The effects of concentration, temperature, and acidity on cellulase activity in the DES/AA-SA solution were investigated to determine the optimum experimental conditions. Residual activity was measured at 6, 12, 24, 48, and 72 h intervals to accurately determine the change in cellulase activity over time. The half-life ($t_{1/2}$) of cellulase under the optimal experimental conditions was then calculated [5]. Each experiment was replicated at least three times to ensure reproducibility.

2.5. Kamlet-Taft parameters

The Kamlet–Taft parameters can be used to describe the properties of DESs. The parameter α represents the ability of a substance to act as an HBD, β represents the ability to accept protons as an HBA, and π* represents the polarizability [34]. In this study, three dyes, Nile red (NR), 4-nitrophenol (OH), and 4-nitroanisole (OMe), were used to determine the Kamlet–Taft parameters.

The α, β, and π* values were calculated using the following formulas:

$$\pi^* = 0.427 \times (34.12 - \upsilon_{OMe}) \quad (2.2)$$

$$\beta = 0.346 \times (35.04 - \upsilon_{OH}) - 0.57\pi^* \quad (2.3)$$

$$\alpha = 19.9657 - 1.0241\pi^* - V_{NR}/1.6078 \quad (2.4)$$

where π* is normalized to cyclohexane and DMSO (Dimethyl sulfoxide), $\upsilon = 1/(\lambda \times 10^{-4})$, and λ is the wavelength corresponding to dye absorption.

2.6. Characterization of DES and cellulase

The DES was analyzed using a Bruker Avance nuclear magnetic analyzer. The solvent was mixed with DMSO in the appropriate amount and tested at 500 MHz. Cellulase and a single enzyme solution were scanned using the F-4600 fluorescence spectrophotometer. The excitation wavelength was 280 nm, the enzyme concentration was 2.5 g/L, and the scanning range was 300–540 nm. The samples were analyzed using a Nicolet-510P Fourier-transform infrared (FTIR) spectrometer. Infrared test analysis was performed using traditional potassium bromide tablets, with an analytical range of 4000–400 cm^{-1}. The enzyme-containing solution was scanned using a Jasco-815 circular dichromate in the range of 190–250 nm. The secondary structure was calculated using CRDATA and SELCON3 software.

2.7. DFT calculation

Quantum calculations were compared with experimental data. One-component structures were optimized using Gaussian16 [35,37] with the DFT B3LYP/6-311G (d,p) procedure, and the dispersion keyword = GD3BJ [36] was added to eliminate dispersion effects. Frequency analysis of the optimized structures was also conducted. The DES structures were constructed using MOPAC [38], and Gaussian16 was employed to generate five random structures for each DES. Optimization was conducted through the B3LYP/6-311G (d,p) procedure, using Molclus [39] for linkage, and subsequently, the molecular structure with the lowest energy from each set was selected for frequency vibrational and density gradient Rendering Dependency Graph (RDG) analyses. Visualization was achieved using Visual Molecular Dynamics (VMD) [40]. The RDG analyses were primarily performed using the Multiwfn program [41].

2.8. MD simulation

The Gromacs2021 [42] MD simulation software was used, and the structure of βG (PDB:3ZYZ) was retrieved from the protein database. The reference solvent used was an AA–SA solution with a pH of 4.8 and a concentration of 0.05 mol. The experimental solvents were ChCl/PEG200 (1:5) and Bet/Gly (1:2) at a concentration of 10 % and 5 %, respectively. The TIP3P model water molecule was selected as the solvent. A temperature of 323 K (50 °C) was adopted in the experiment. To simulate the DES/AA-SA solution environment, HBA and HBD molecules were added according to the AA–SA solvent environment [43]. The box dimensions in the x, y, and z directions met periodic boundary conditions, and the system was electrically neutral [44]. Optimized Potentials for Liquid Simulations-All Atom (OP-LS) was used as the force field [45].

Prior to MD simulation, the energy of the system was minimized via the steepest descent method. A two-step pre-balancing simulation was performed. First, the system was balanced for 1 ns under the Isothermal-isovolumetric (NVT) ensemble to reach the target temperature of 323 K. Second, the density of the isothermal–isobaric (NPT) ensemble was balanced for 1 ns according to the previous step. Finally, a 100 ns production-phase simulation was conducted under the NPT system, with all position restrictions lifted. The particle mesh Ewald (PME) method was used throughout all stages to manage long-range electrostatic actions [46]. For proximity search, the Verlet [47] method was employed. The Coulomb truncation distance was set to 1.0 nm, and the Lennard-Jones (LJ) or Buckingham threshold was set to 1.0 nm. The V-rescale method was used for temperature coupling during NVT equilibrium, and

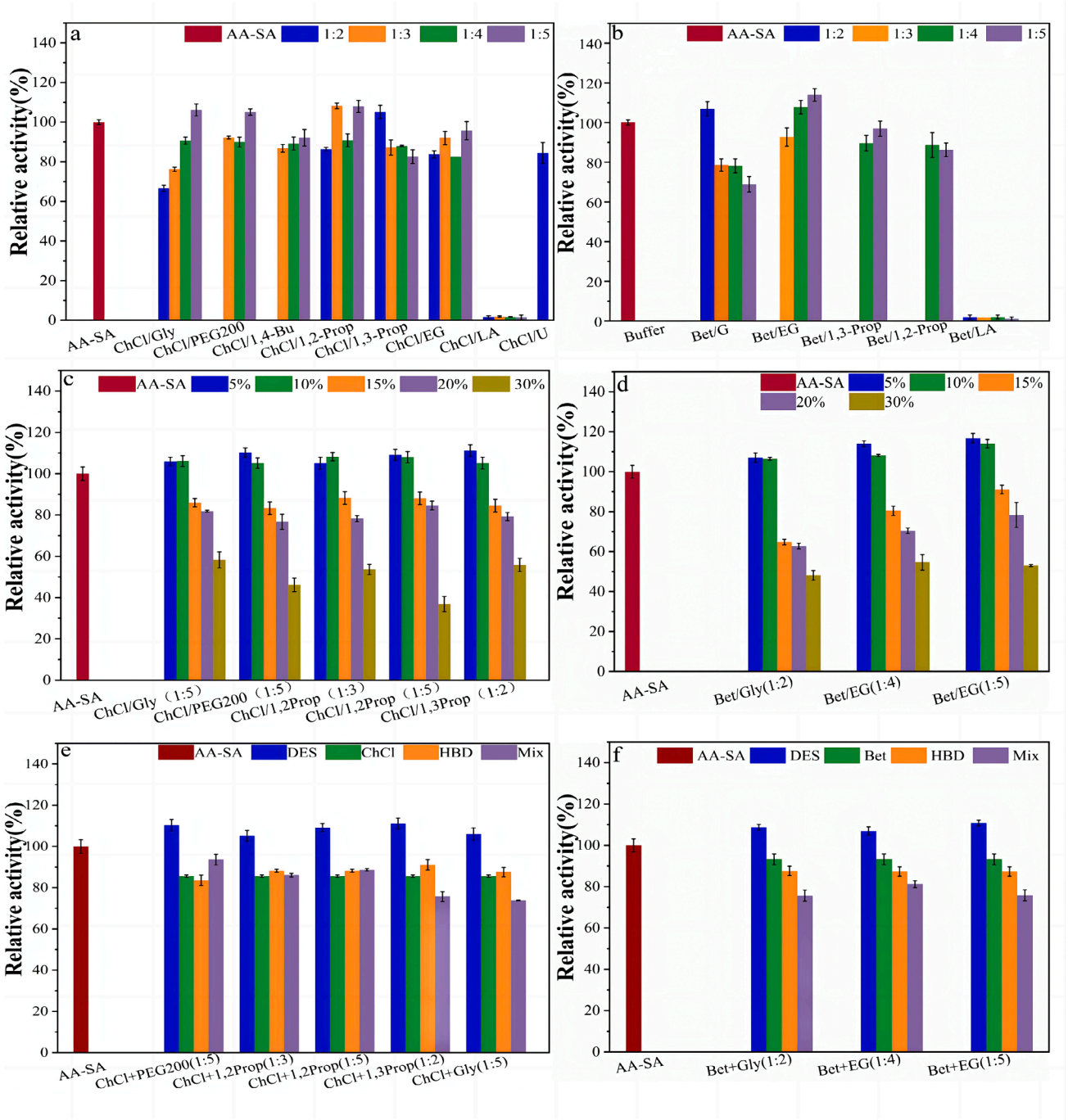


Fig. 1. The relative activity of cellulase in ChCl-DES/AA-SA (a) and Bet-DES/AA-SA (b) solutions containing specific ratios of HBDs at a concentration of 10% (v/v). The effects of concentrations of ChCl-DESs (c) and Bet-DESs (d) on cellulase activity. The effects of components of ChCl-DESs (e) and Bet-DESs (f) on cellulase activity at a concentration of 5% (w/v). The relative activity = (FPA in DES/AA-SA solution / FPA in AA-SA solution) * 100%.

the coupling reference temperature was set to 323 K. During the NPT equilibrium and production-phase simulation, the Berendsen method was used for pressure coupling, and the coupling reference pressure was set to 1 atm.

2.9. Enzymatic hydrolysis of biomass in DES/AA-SA solution

An appropriate amount of corn stover or poplar wood was mixed with DES (ChCl/EG (1:5)) in a ratio of 1:30, put into a high-pressure reactor and the reaction was carried out in a homogeneous reactor at a reaction temperature of 200 °C for 3 h. After the reaction was finished, the solid substrate was filtered and washed to be neutral. Subsequently, the solid substrate was freeze-dried for 48 h and stored at 4 °C, ready for use in cellulase-catalyzed hydrolysis [48].

1 g of pretreated substrate was added to a solution containing 50 mL of AA–SA, 10 % ChCl-DES/AA-SA solutions or 5 % Bet-DES/AA-SA solutions. The mixture was then hydrolyzed in a thermostatic generator at 50°C and 150 rpm for 96 h using 7.5 FPU/g-substrate of cellulase dosage and 30 U/g-substrate of βG dosage. Samples were taken at 1, 3, 6, 9, 12, 24, 36, 48, 72 and 96 h. The glucose content was determined by HPLC using a Shimadzu LC-20AD, Aminex HPX-87H column at 65 °C and isocratic elution with 5 mM H_2SO_4 mobile phase. The glucose yield was calculated using the following formula:

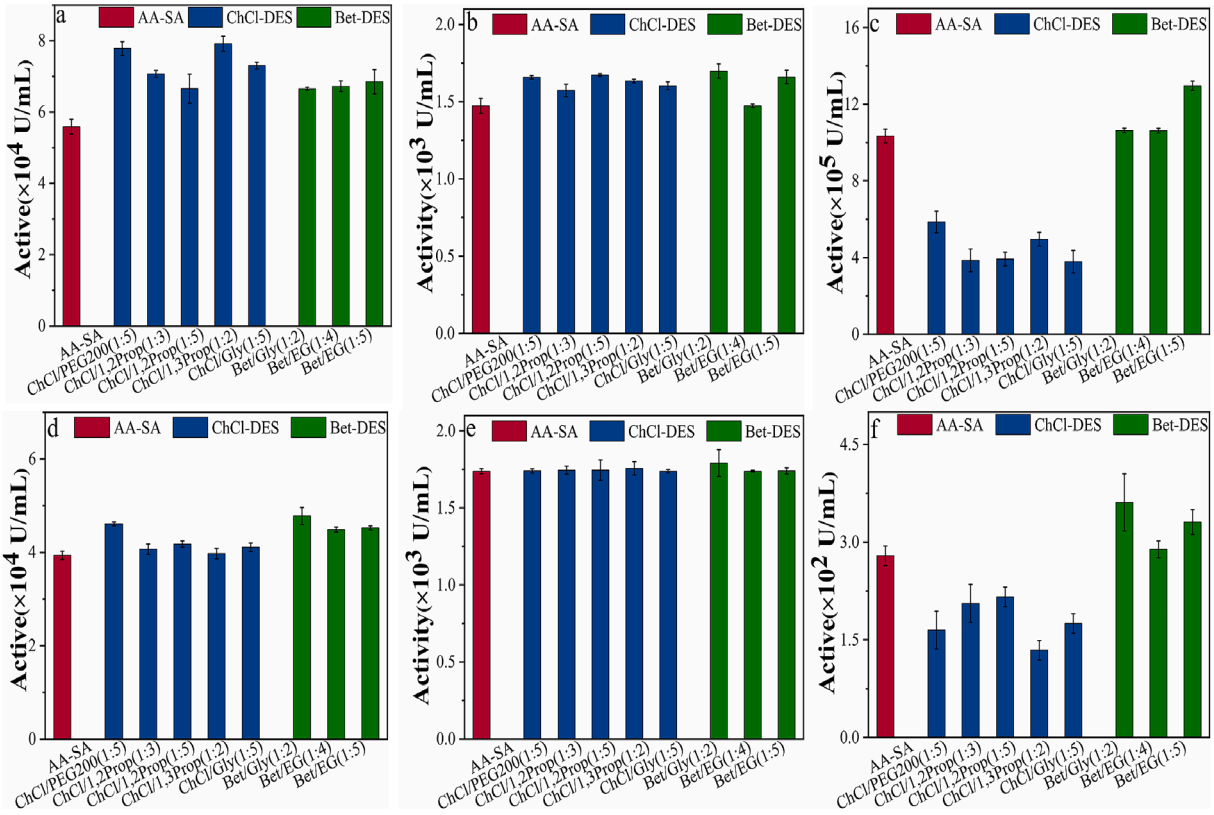


Fig. 2. The β-glucosidase (a), *endo*-glucosidase (b) and *exo*-glucosidase (c) activities of cellulase, the activity of individual β-glucosidases (d), and the *endo*-glucosidase (e) and *exo*-glucosidase (f) activities of pulping enzyme in DES/AA-SA solutions with a DES concentration of 5% (w/v).

$$Glucoseyield(\%) = \frac{Glucoseproducedbyenzymatichydrolysis(g)}{Totalglucaninsubstrate(g) \times 1.1} \times 100 \quad (2.5)$$

The unproductive adsorption of residue lignin on cellulase was performed. The residual lignin was prepared according to the method described by Nakagame [49], and the lignin obtained was freeze-dried for use. The lignin adsorption experiment was carried out as described in references [50,51]. The resulting lignin (0.5 g) was dispersed in 50 mL of AA-SA, 10 % (v/v) ChCl-DES/AA-SA or 5 % (v/v) Bet-DES/AA-SA solution, corresponding to a lignin content of 1 % (w/v). The cellulase was then added at a dose of 15 FPU/g lignin. Samples were taken at 1, 3, 6, 9, 12, 24, 36, 48, 72, and 96 h, then centrifuged and filtered using a water filter. The protein content of the samples was determined using the Coomassie Brilliant Blue reagent, and calculated using the appropriate formula [52].

3. Results and discussions

3.1. Construction of a DES-based cellulase hydrolysis system

The relative activity was used to better assess the effect of DES on cellulase activity. If the relative activity is greater than 100 %, it indicates that the DES has a promoting effect on cellulase activity; if the relative activity is less than 100 %, it indicates that the DES has an inhibiting effect on cellulase activity. ChCl and Bet were used as HBAs in the synthesis of two forms of DESs. The ChCl- and Bet-based DESs were formed using a variety of HBDs from the alcohol, acid, and base families in different molar ratios. The preparation yielded approximately 45 DESs that retained their liquid state at room temperature (Table S1). The DESs that exhibited thermal stability at room temperature were then used to generate mixtures with a conventional AA–SA solution (pH 4.8, 0.05 M). The resulting DES/AA-SA solutions, namely ChCl-DES/AA-SA and Bet-DES/AA-SA, were used as a novel hydrolysis system for cellulase.

3.1.1. The enzyme activities in DES/AA-SA solutions

The incorporation of ChCl-DESs (Fig. 1a) and Bet-DESs (Fig. 1b) containing specific ratios of alcohol HBDs showed an enhancement in cellulase activity. Among ChCl-DESs, ChCl/Gly (1:5), ChCl/PEG200 (1:5), ChCl/1,2-prop (1:3), ChCl/1,2-prop (1:5), and ChCl/1,3-prop (1:2) improved cellulase activity. Among Bet-DESs, Bet/Gly (1:2), Bet/EG (1:4), and Bet/EG (1:5) enhanced cellulase activity. In contrast, the introduction of DESs with acidic and basic HBDs was detrimental to enzyme performance (Fig. 1a,b), with cellulase showing negligible activity in DESs with an acidic HBD.

The optimal DES concentration in the DES/AA-SA solutions was 5 % or 10 % (v/v), resulting in higher cellulase activity compared with the conventional AA–SA solution (Fig. 1c,d). As shown in Table S2, the cellulase activity values at 5 % and 10 % (v/v) DES concentrations were higher than that in AA–SA solution. The cellulase activity values at 5 % (v/v) DES were slightly higher than those at 10 % (v/v) DES. However, when the DES concentration exceeded 10 % (v/v), the cellulase activity decreased significantly and fell well below the initial value in AA-SA solution. The high concentration of DES leads to high viscosity, which hinders mass transfer during the reaction [28]. Fine-tuning of conformational flexibility is crucial for enzyme catalysis [53]. At a high concentration, more DESs will interact with the enzymes through hydrogen bonding, which could limit the conformational flexibility of the enzyme [54]. Therefore, the above two possible reasons explain why the cellulase activity was slightly higher at 5 % (v/v) DES compared to 10 % (v/v) DES and why the reduction in activity becomes more pronounced at much higher concentrations of DESs. Specifically, at the 5 % (v/v) DES concentration, cellulase activity increased by 5.9 % for ChCl/Gly (1:5), 10.2 % for ChCl/PEG200 (1:5), 5.1 % for ChCl/1,2-prop (1:3), 9.1 % for ChCl/1,2-prop (1:5), 11.1 % for ChCl/1,3-prop (1:2), 6.9 % for Bet/Gly (1:2), 7.7 % for Bet/EG (1:4), and 13.9 % for Bet/EG (1:5).

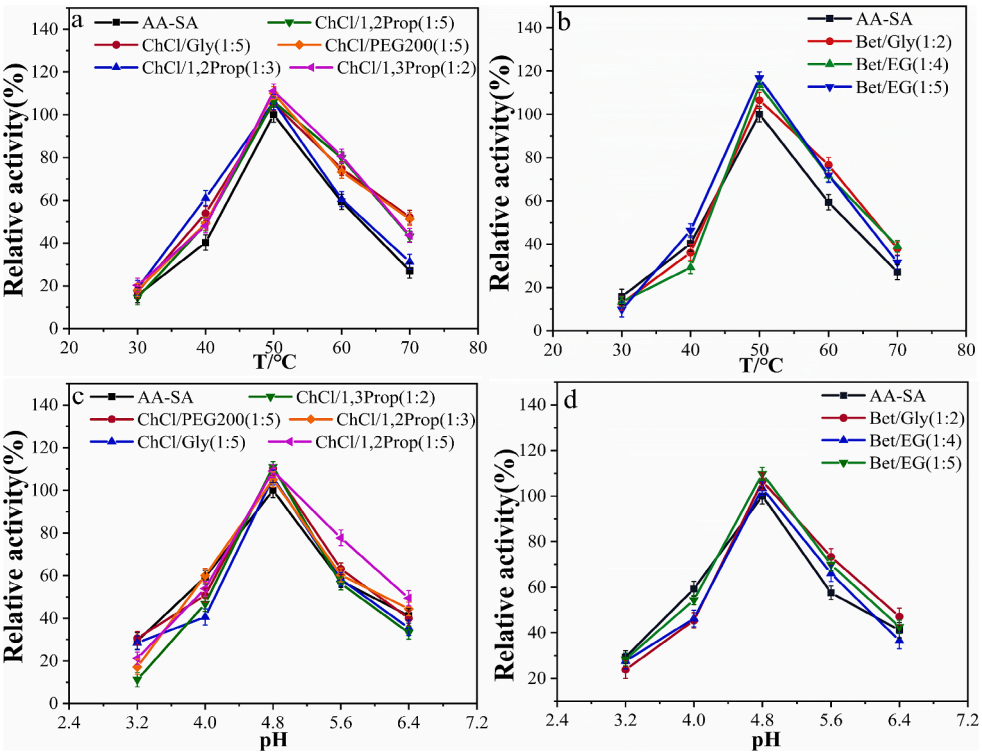


Fig. 3. The effects of temperature (a,b) and pH (c,d) on cellulase activity in DES/AA-SA solutions.

To determine whether the increase in cellulase activity was induced by the synthesized DES or its components, the effects of DESs and their components on cellulase activity were compared, as shown in Fig. 1(e,f). AA–SA solutions containing both ChCl-DESs and Bet-DESs exhibited significantly higher cellulase activities than those containing the individual components or mixtures. The cellulase activities in AA–SA solutions containing single DES components or mixtures were lower than the baseline value in AA-SA solution. This suggests that the improved cellulase activity was due to the unique properties of the synthesized DES, rather than merely the mixing of its components.

Cellulase is a multi-component enzyme complex that works synergistically to break down cellulose. The main enzyme components [55,56] are *endo*-glucosidase (EG), which randomly cleaves internal glycosidic bonds in cellulose chains at random; *exo*-glucosidase/cellobiohydrolase (CBH), which cleaves cellobiose units from cellulose chain ends; and βG, which breaks down cellobiose into glucose. At the optimal DES concentration of 5 % (v/v), the effects of DES on the activities of these three enzyme components in commercial cellulase was investigated, as shown in Fig. 2(a-c). To avoid the interference between different components, the individual βG was purchased and the effects of DESs on its activity was also investigated, as shown in Fig. 2d. Commercial EG and CBH enzymes are typically mixtures of the two, which are commonly used in the pulp and paper industry, referred to as pulping enzyme. Hence, we also studied the effects of DESs on the EG activity (Fig. 2e) and CBH activity (Fig. 2f) of pulping enzyme.

Table 1
The half-life of cellulase after the addition of DESs in AA-SA solutions with concentrations of 5% (v/v) and 10% (v/v).

DES	Concentration (v/v %)	Half-life (h)	Concentration (v/v %)	Half-life (h)
AA-SA	–	28.2	–	28.2
ChCl/PEG200 (1:5)	10	33.8	5	33.7
ChCl/1,2-prop (1:3)	10	45.9	5	39.4
ChCl/1,2-prop (1:5)	10	40.5	5	35.4
ChCl/1,3-prop (1:2)	10	95.2	5	81.8
ChCl/Gly (1:5)	10	51.3	5	45.6
Bet/Gly (1:2)	10	66.2	5	86.6
Bet/EG (1:4)	10	40.3	5	48.8
Bet/EG (1:5)	10	36.1	5	41.5

At a DES concentration of 5 % (v/v), the βG activities of cellulase were improved by 20 %–35 % after the addition of ChCl-DESs and Bet-DESs to the AA–SA solution (Fig. 2a), while the EG activities of cellulase showed a slight increase of 10 %–15 % (Fig. 2b). The CBH activities of cellulase were significantly reduced in ChCl-DES/AA-SA solutions compared with the AA–SA solution, whereas they were slightly improved in Bet-DES/AA-SA solutions (Fig. 2c). Moreover, the individual βG activity was observed to be improved in both ChCl-DES/AA-SA and Bet-DES/AA-SA solutions (Fig. 2d). The pulping enzyme, which is a complex of EG and CBH, displayed similar trends in EG and CBH activities as cellulase. Specifically, the EG activity of the pulping enzyme in either ChCl-DES/AA-SA or Bet-DES/AA-SA solutions showed no significant changes compared with that in the AA–SA solution (Fig. 2e), while the CBH activity decreased in ChCl-DES/AA-SA solutions but increased in Bet-DES/AA-SA solutions (Fig. 2f). Cleary, the improved cellulase activity is mainly due to the increase in βG and EG activities, especially the improvement in βG activity. The βG activity values in commercial cellulase and individual βG enzymes under different DES compositions and concentrations are shown in Tables S3 and S4, respectively. The βG activity values at 5 % and 10 % (v/v) DES concentration were higher than that in AA–SA solution, with the values at 5 % being greater. Furthermore, when the DES concentration exceeded 10 %, the βG activity values were lower than those in AA-SA solution. Among ChCl-DESs, ChCl/PEG200 (1:5) showed a good improvement in βG activity for both enzymes. Among Bet-DESs, Bet/Gly (1:2) showed a good improvement in βG activity for both enzymes. The significant increase in βG activity could reduce the rapid accumulation of cellobiose during the cellulose hydrolysis process, alleviating cellobiose inhibition on the

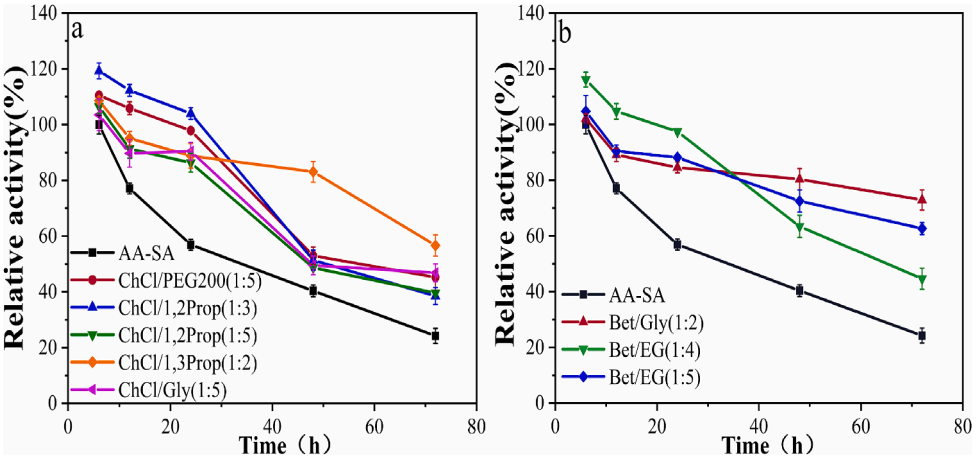


Fig. 4. The cellulase activity in ChCl-DES/AA-SA (a) and Bet-DES/AA-SA (b) solutions over 72 h.

enzymes and improving overall efficiency [57,58].

The optimal reaction conditions for cellulase in the DES/AA-SA solutions were investigated by plotting the activity at different temperatures and pH values (Fig. 3). Similar to the AA–SA solution, the optimal temperature and pH for cellulase in the DES/AA-SA solutions were approximately 50 °C and pH 4.8. Within the tested temperature range of 30 °C–70 °C, DES/AA-SA solutions exhibited higher cellulase activities than the AA–SA solution (Fig. 3a,b). Under highly acidic conditions (pH $\leq$ 4.0), the cellulase activity in AA-SA solution was higher than those in DES/AA-SA solutions, but at mild acidity (pH $\geq$ 4.8) cellulase activities in DES/AA-SA solutions were higher (Fig. 3c,d). This indicates that the cellulase in DES/AA-SA solutions exhibited good thermal stability and weak acid stability, but not strong acid resistance.

3.1.2. *The stability of cellulase in DES/AA-SA solutions*

The half-life of enzymes is important in assessing enzyme stability and applicability [28]. Table 1 shows the half-life of cellulase in different solutions. In the conventional AA–SA solution, the half-life of cellulase was 28 h; that is, the time for the enzyme activity to decrease by half was 28 h. In different DES/AA-SA solutions, the half-life of cellulase increased to various extents, indicating that the novel DES/AA-SA systems improved cellulase stability. The concentration of DES also had an important effect on the half-life of cellulase. In ChCl-DES/AA-SA solutions, the half-life of cellulase at a concentration of 10 % (v/v) was higher than that at a 5 % (v/v) concentration (Table 1). Among the DESs, the introduction of ChCl/1,3-prop (1:2) resulted in the most significant improvement in the half-life of cellulase, with the half-life extending to 95 h at a 10 % concentration, ~3.4 times the initial value. In the Bet-DES/AA-SA solutions, the half-life of cellulase at 5 % (v/v) concentration was higher than that at 10 % (v/v). The addition of Bet/Gly (1:2) yielded the most significant improvement in the half-life of cellulase, which reached 87 h at a 5 % (v/v) concentration, ~3.1 times the initial value. The significantly prolonged enzyme half-life is important for industrial applications, as it means that the enzyme can maintain its activity for a longer time, enabling longer reactions or processes. Yao et al. [28] investigated the effect of DESs on the half-life of cellulase, and an improvement was also observed with specific DESs. When the same Bet/Gly (1:2) was used, a similar degree of improvement was found, and the half-life was approximately three times the initial value. Improvements in the half-life of other enzymes such as laccase, cutinase, epoxide hydrolase and lipase by DES have also been reported, with half-life increases of 2–10 times [17,59–61]. In addition, different DESs had different effects on improving the half-life of cellulase, which is consistent with our study. This indicates that the structural characteristics of DES have an important effect on the cellulase stability.

Considering the higher improvement in the half-life of cellulase with 10 % (v/v) ChCl-DES and 5 % (v/v) Bet-DES (Table 1), the half-life of individual βG was also investigated at the corresponding concentrations, as shown in Table S5. Among ChCl-DESs (10 %, v/v), ChCl/Gly (1:5) showed the highest improvement in βG half-life with a value of 66.1 h. ChCl/PEG200 (1:5) also showed a significant improvement in βG half-life with a value of 51.4 h. Among Bet-DESs (5 %, v/v), Bet/Gly (1:2) showed the highest improvement in βG half-life with a value of 45.5 h. Combined with the results shown in Fig. 2, it can be seen that both ChCl/PEG200 (1:5) and Bet/Gly (1:2) not only significantly improved the activity of βG but also prolonged its half-life, demonstrating a clear dual advantage in improving both the stability and βG activity.

To further assess the stability of cellulase in DES/AA-SA solutions, changes in cellulase activity over 72 h were examined (Fig. 4). Cellulase activity in the AA–SA solution decreased sharply over time, falling to < 60 % after 24 h and ~ 30 % after 72 h. In contrast, the activity in both ChCl-based and Bet-based DES/AA-SA solutions decreased slowly within the first 24 h and remained at least 80 %. After 24 h, the decrease in activity varied according to the type of DES. In particular, ChCl/1,3-prop (1:2) at 10 % (v/v) and Bet/Gly (1:2) at 5 % (v/v), which were characterized by longer cellulase half-lives, showed a lower decline after 24 h. Under 10 % (v/v) ChCl/1,3-prop (1:2), the activity remained above 80 % at 48 h, and above 40 % at 72 h. Under 5 % (v/v) Bet/Gly (1:2), the activity remained around 80 % even after 72 h. The results indicate that the DES/AA-SA solutions enabled the cellulase to maintain excellent stability over long periods.

3.2. *Characterization of enzymes*

3.2.1. *The fluorescence spectra of enzymes*

Fluorescence spectroscopy was employed to study the effects of DESs on the conformation of enzymes (Fig. 5). Protein fluorescence mainly originates from three aromatic amino acid residues: tryptophan (Trp), tyrosine (Tyr), and phenylalanine (Phe) [62,63]. Typically, the fluorescence intensity ratio of these residues is 100:9:0.5, indicating that Trp residues are the major contributors to protein fluorescence [64]. Fig. 5 illustrates the presence of fluorescence absorption peaks around 350 nm for cellulase (Fig. 5a,b), individual βG (Fig. 5c,d), and pulping enzyme (Fig. 5e,f). All enzymes in both ChCl-DES/AA-SA and Bet-DES/AA-SA solutions exhibited an enhancement of fluorescence. This is attributable to the hydrogen bonding interactions of the DES, which stabilize the spatial conformations of enzyme molecules and lead to compact structures, thereby limiting conformation mobility. Consequently, enhanced fluorescence emission occurs, similar to the aggregation-induced emission effects documented in the literature [65].

In addition to fluorescence enhancement, the introduction of DESs also resulted in a blueshift of the maximum emission peak (λ_{max}) for

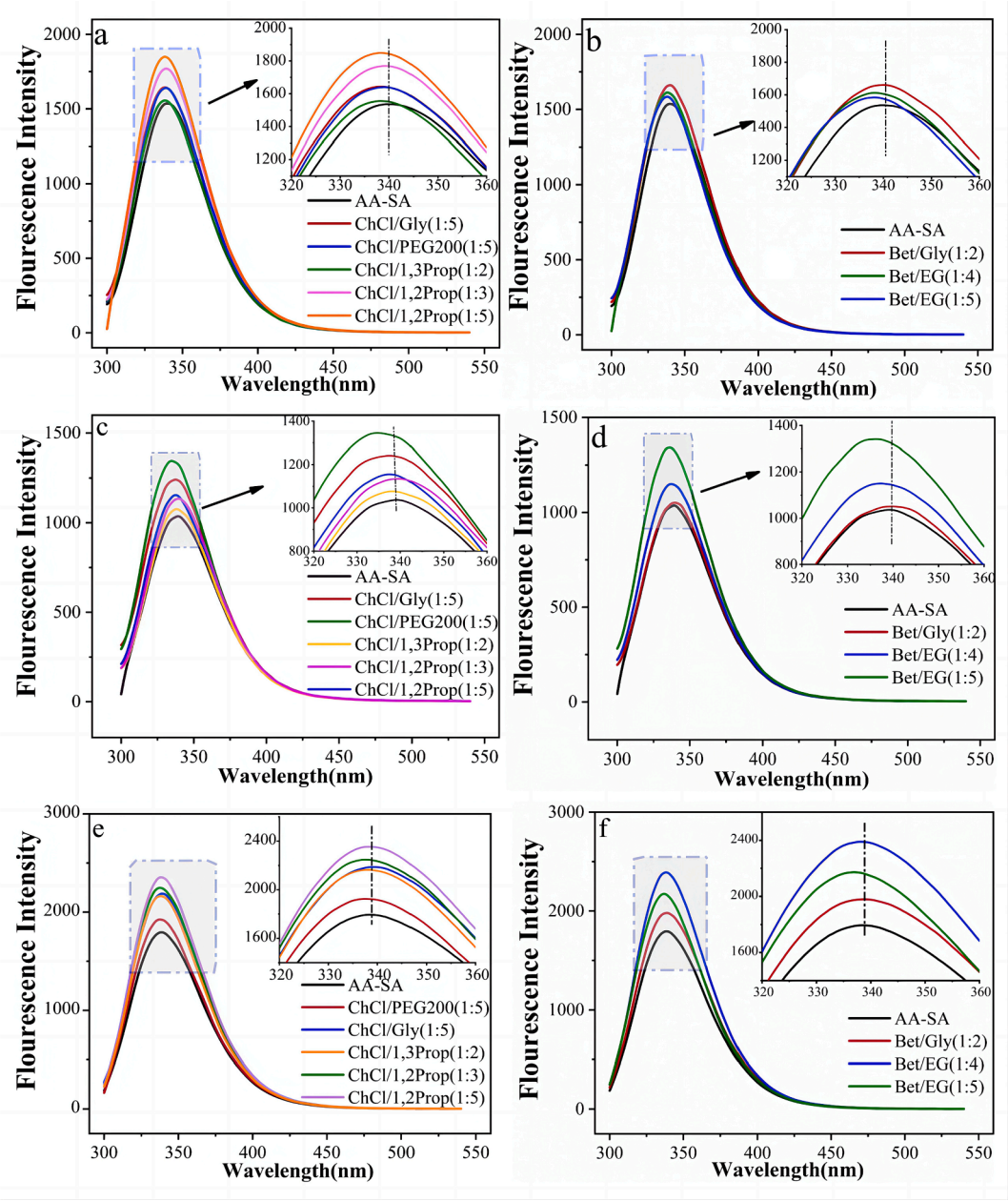


Fig. 5. The fluorescence spectra of cellulase (a,b), individual β-glucosidase (c,d), and pulping enzyme (e,f) in DES/AA-SA solutions.

cellulase (Fig. 5a,b) and βG (Fig. 5c,d). In contrast, the λ_{max} of the pulping enzyme (Fig. 5e,f) did not show any significant change. This indicates that the blueshift observed for the cellulase was driven by the blueshift related to βG, which is an important component of the cellulase. The degree of blueshift in λ_{max} for Trp residues can indicate the extent of change in the spatial structure of the enzyme [66]. The order of blueshift observed for cellulase from highest to lowest was ChCl/1,3-prop (1:2) (3.2 nm) > Bet/EG (1:5) (3 nm) > ChCl/PEG200 (1:5) = ChCl/1,2-prop (1:3) (2.2 nm) > Bet/EG (1:4) (2 nm) > ChCl/1,2-prop (1:5) (1.2 nm) > ChCl/Gly (1:5) (1 nm) > Betl/Gly (1:2) (0.6 nm). The order of blueshift observed for βG was Bet/EG (1:5) (5.6 nm) > ChCl/1,2-prop (1:3) (4.8 nm) > ChCl/Gly (1:5) (2.6 nm) > Bet/EG (1:4) (2.4 nm) > ChCl/PEG200 (1:5) (2 nm) > ChCl/1,3-prop (1:2) (1.2 nm) > ChCl/1,2-prop (1:5) (0.6 nm) > Betl/Gly (1:2) (0.2 nm).

As evidenced by the improvement in cellulase and βG activities, the alteration in the spatial position of the Trp residues is advantageous for the function of the βG activity, consequently enhancing the cellulase activity. However, the DES inducing the maximum and minimum blueshift did not correspond to the maximum cellulase and βG activities. This suggests that appropriate changes in spatial conformation are more conducive to achieving maximum enzyme activity. Research has shown that Trp residues are crucial for the catalytic function of both EG and CBH [67,68]. However, DES addition did not significantly alter the spatial positions of the Trp residues for EG and CBH in the pulping enzyme (Fig. 5e,f), resulting in minimal changes in their activities. ChCl-DES addition may even reduce the activity of CBH.

3.2.2. *The circular dichroism spectra of cellulase*

Circular dichroism (CD) was employed to assess the secondary structure contents (α-helix, β-sheet, β-turn, and random coil) of cellulase in conventional AA–SA solutions, ChCl-DES/AA-SA solutions, and Bet-DES/AA-SA solutions (Fig. 6). The CD spectra of the cellulase in AA–SA solutions showed a broad and large negative peak at ~ 210 nm, indicating a tertiary structure with "α + β" configuration [69]. The CD

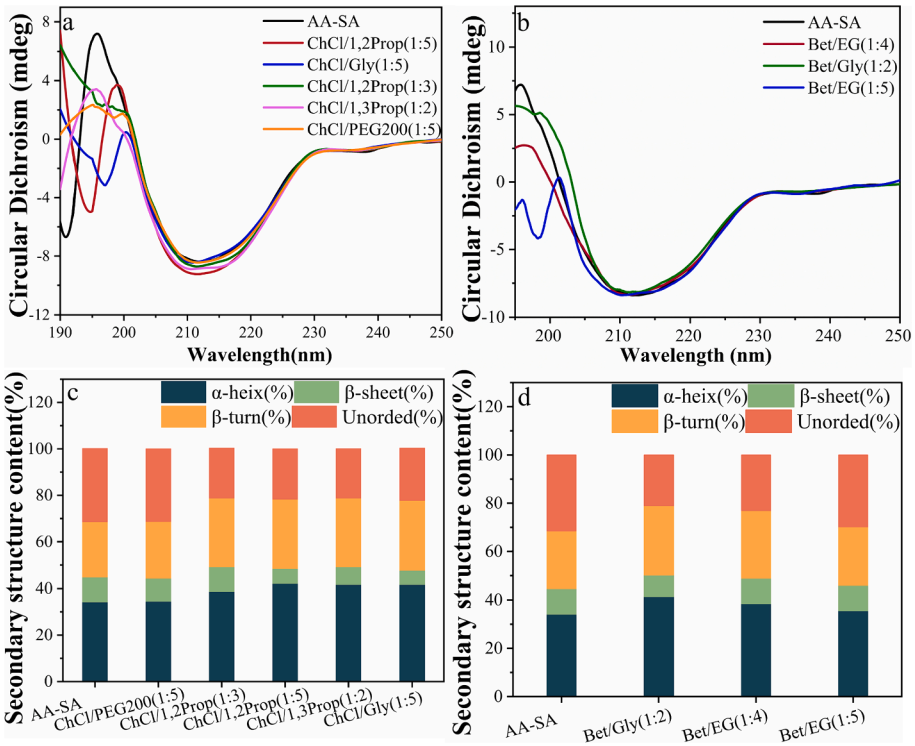


Fig. 6. The circular dichroism spectra of cellulase in ChCl-DES/AA-SA (a) and Bet-DES/AA-SA (b)solutions and the relative content of secondary structure for cellulase in ChCl-DES/AA-SA (c) and Bet-DES/AA-SA (d)solutions.

curves of the cellulase in the ChCl-DES/AA-SA (Fig. 6a) and Bet-DES/AA-SA (Fig. 6b) solutions were broader and more pronounced than the curves for the AA–SA solutions. This indicates that the addition of DESs to the AA–SA solution altered the relative proportion of secondary structure in cellulase, resulting in a more compact spatial conformation of enzymes [70]. The specific content of secondary structures for cellulase in different solutions was calculated using CRDATA.EXE and SLECON3.EXE, as displayed in Fig. 6. Introducing the ChCl-DESs (Fig. 6c) and Bet-DESs (Fig. 6d) increased the relative contents of α-helix and β-turn and reduced the relative contents of β-sheet and unordered structure. The changes in the secondary structure contents of βG and pulping enzyme were determined according to the area of the amide I band (1600–1700 cm^{-1}) in the FTIR spectra (Fig. S1). The results also showed a significant increase in the contents of α-helix and β-turn after the incorporation of ChCl-DES and Bet-DES, while the contents of β-sheet and unordered structure significantly decreased.

3.3. Characterization of DES

3.3.1. The ^1H NMR spectra of DES

Hydrogen bonding between HBDs and HBAs is the driving force behind the formation of DESs [71]. The hydrogen bonds can induce chemical shifts in the hydroxyl protons of HBDs, as observed in ^1H NMR spectra. The mechanism involves a reduction in electron density around the hydroxyl hydrogens upon hydrogen bond formation, causing the resonance absorption of the hydroxyl protons to shift to a lower field and increase the chemical shift [72]. This effect is evident in the ^1H NMR spectra of the resulting DESs (Fig. 7). In the ChCl/Gly (1:5), the hydroxyl proton peak of Gly shifted from 4.42 to 4.50 ppm (Fig. 7a). Similarly, in the ChCl/PEG200 (1:5), the hydroxyl proton peak of PEG200 shifted from 4.56 to 4.59 ppm (Fig. 7b). In addition, in the ChCl/1,3-prop (1:2), the hydroxyl peak of 1,3-prop shifted from 4.30 to 4.33 ppm (Fig. 7c). In the ChCl/1,2-prop (1:3) and ChCl/1,2-prop (1:5), the hydroxyl peak of 1,2-prop shifted from 4.42 to 4.49 (1:3) and 4.46 ppm (1:5), respectively (Fig. 7d). The Bet/Gly (1:2) showed a shift of the hydroxyl proton peak of Gly from 4.42 to 4.48 ppm (Fig. 7e). Interestingly, the hydroxyl peaks of EG in the Bet/EG (1:4) and Bet/EG (1:5) DES systems shifted to a higher field, from 4.47 to 4.44 ppm (Fig. 7f), possibly due to more intramolecular hydrogen bonding and greater hydroxyl proton electron density due to increased EG content, thereby reducing the chemical shift. In the optimal hydrolysis solutions of ChCl-DES/AA-SA (10 %) and Bet-DES/AA-SA (5 %) the hydroxyl absorption peaks of the HBDs can still be observed in the 4–6 ppm range. Despite weakened peak intensities due to dilution effects, shifts induced by hydrogen bonding persisted, indicating that hydrogen bonding was preserved at these DES concentrations.

3.3.2. The Kamlet–Taft values of DESs

The hydrogen bonding property of DESs can be characterized by Kamlet–Taft parameters [73]. The Kamlet–Taft system describes solvent polarity using three parameters: α, β, and π*. α indicates the hydrogen bond-donating ability, β indicates the hydrogen bond-accepting ability, and π* indicates the polarity. Among ChCl-DESs (Fig. 8a–c), the α values ranged from 0.87 to 1.28, and the β values ranged from 0.69 to 0.97. The ChCl/1,2-prop (1:3) with an α value of 1.03 and a β value of 0.94 showed the highest improvement in cellulase activity, with an 8 % increase (Fig. 1). Among Bet-DESs (Fig. 8e–f), the α values ranged from 1.10 to 1.25, and the β values ranged from 0.74 to 0.93. The Bet/EG (1:5) DES with an α value of 1.10 and a β value of 0.85 exhibited the greatest improvement in cellulase activity, with a 14 % increase (Fig. 1).

The amphiprotic nature of proteins allows them to form hydrogen bonds with DESs by either donating or accepting protons. The interaction between DESs and enzymes could be modulated by the α and β values of the DESs. DESs with smaller α and β values exhibit weaker hydrogen bonding ability, leading to less binding to enzyme molecules; consequently, they induce minor changes in the spatial conformation of the enzyme, resulting in limited improvement in cellulase activity. Conversely, DESs with larger α and β values have stronger hydrogen bonding ability, which enhances the binding of the DESs to enzymes and induces greater changes in spatial conformation, resulting in less

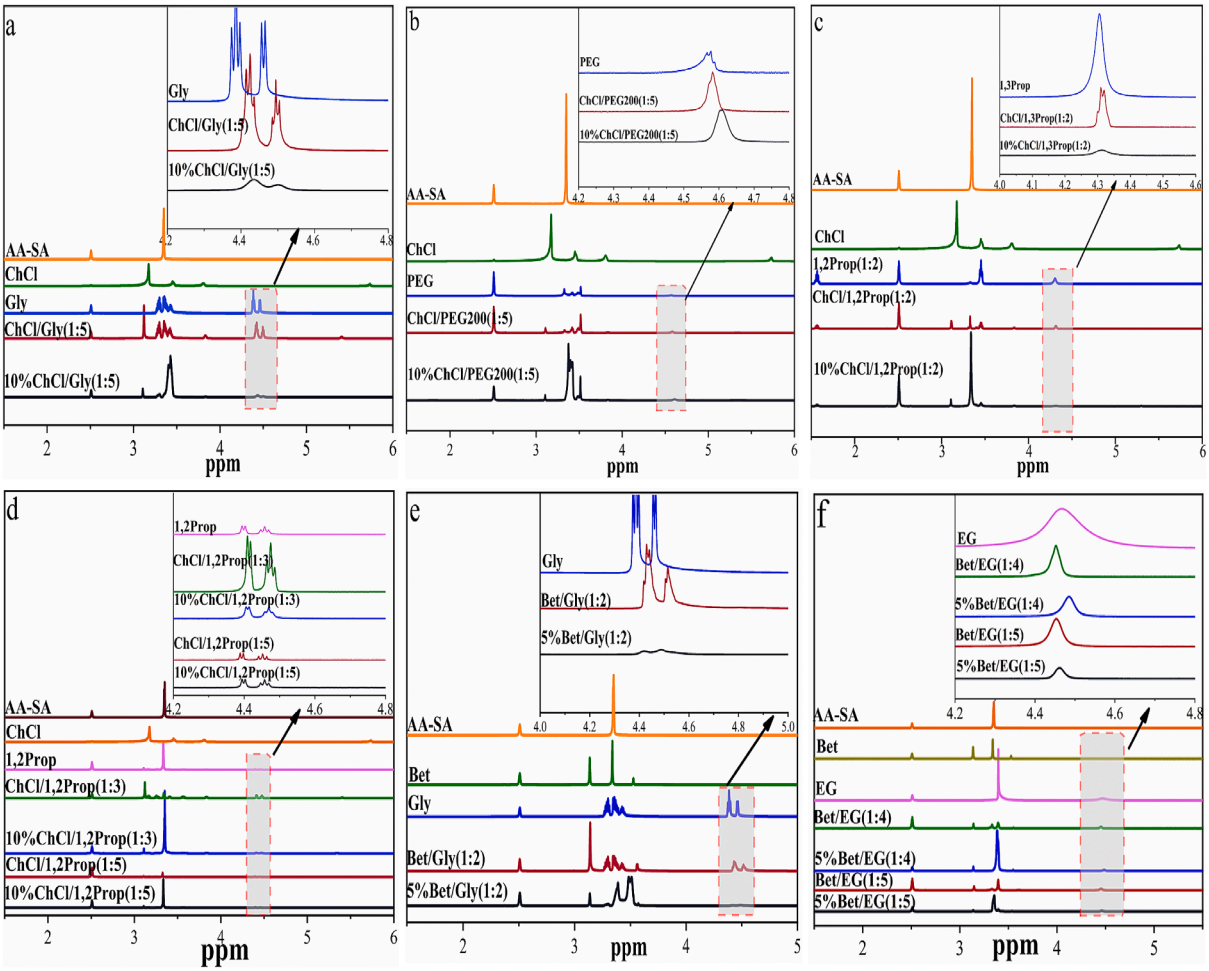


Fig. 7. ^{1}H NMR spectra of HBAs, HBDs, and the synthesized DESs.

improvement in cellulase activity. Thus, appropriate α and β values facilitate moderate hydrogen bonding between DESs and cellulase, promote reasonable conformational changes, and significantly improve cellulase activity. Interestingly, the improvement in cellulase activity was strongly negatively correlated with the polarity of the DES (Fig. 8c, f). Increased polarity enhances the interactions between DESs and cellulase, particularly in terms of the spatial distribution of Trp residues, potentially hindering enzyme conformational optimization and substrate contact, thereby reducing the improvement effect.

3.3.3. *The DFT calculation of DESs*

The most stable DES structures with the lowest energy were obtained through DFT calculation (Fig. 9a). The DESs exhibited different types of hydrogen bonds with lengths ranging from 1.65 to 3. ChCl/1,3-prop (1:2) exhibited bonds FC-H·Cl (2.51), OH-H·Cl (2.02), and OH-H-HO (1.77). ChCl/1,2-prop (1:3) exhibited bonds FC-H·Cl (2.44), OH-H·Cl (1.98), and OH-H-HO (1.68). ChCl/1,2-prop (1:5) exhibited bonds OH-H·Cl (2.09), OH-H-HCF (1.81), OH-HHOCF (1.73), and OH-H-OCF (1.40). ChCl/Gly (1:5) exhibited bonds OH-H·Cl (2.51) and OH-HHOCF (2.16). ChCl/PEG2001:5 exhibited bonds OH-H·Cl (4.28), OH-HHO-H·Cl (2.65), and OH-HHOCF (1.77). Bet/Gly (1:2) exhibited bonds OH-HHOCF (1.83) and OH-H-OCF (1.77). Bet/EG (1:4) exhibited bonds OH-HHOCF (1.81) and OH-H-OCF (1.81). Bet/EG (1:5) exhibited bonds OH-HHOCF (1.87) and OH-H-OCF (1.81). In addition, the optimal structure was analyzed using a rendering dependency graph (RDG), as shown in Fig. 9b. According to the value of the symbol $(_2\lambda)\rho(r)$, the surface was colored using a red–blue–green scale: blue indicates strong attractive interactions (such as hydrogen bonds and halogen bonds), red indicates strong repulsion, and the transition zone indicates typical van der Waals interactions. According to Fig. 9b, the DESs exhibited equipotential surfaces, which are typical of van der Waals interactions. This indicates that in addition to hydrogen bond interactions, weak van der Waals forces existed between the HBD and the HBA.

The analysis of HOMO and LUMO energy levels (Fig. 9c) provided insights into the structural stabilities of DESs. A smaller HOMO–LUMO energy gap indicates easier electron transitions and a more stable structure. The selected DESs exhibited relatively low energy gaps of approximately 6–7 eV, indicating stable systems. Among ChCl-DESs, the energy gap variation followed this trend: ChCl/1,3-prop (1:2) > ChCl/Gly (1:5) > ChCl/1,2-prop (1:3) > ChCl/1,2-prop (1:5) > ChCl/PEG200 (1:5). Among Bet-DESs, the variation of the energy gap follows this trend: Bet/Gly (1:2) > Bet/EG (1:4) > Bet/EG (1:5). Notably, a strong negative linear relationship existed between the energy gap of the DES and the half-life of the cellulase (Fig. 9d,e). Lower energy gaps in the DES corresponded to higher stability and a longer half-life of cellulase in the DES/AA-SA solution. Among the ChCl-DESs, ChCl/1,3-prop (1:2) exhibited the lowest energy gap of 6.09 eV, indicating the highest stability. Correspondingly, the half-life of cellulase in the ChCl/1,3-prop (1:2)/AA-SA solution reached the highest value of 95.2 h, 3.4 times the baseline. Among the Bet-DESs, Bet/Gly (1:2) had the lowest energy gap of 6.09 eV, indicating the highest stability. In the Bet/Gly (1:2)/AA-SA solution, the half-life of cellulase reached the highest value of 86.6 h, 3.1 times the baseline value. The encapsulating effect of DESs on enzymes compacts their conformation, reducing their mobility. This allows the enzymes to withstand external perturbations to their conformation for an extended period, thereby increasing their half-life. DESs with

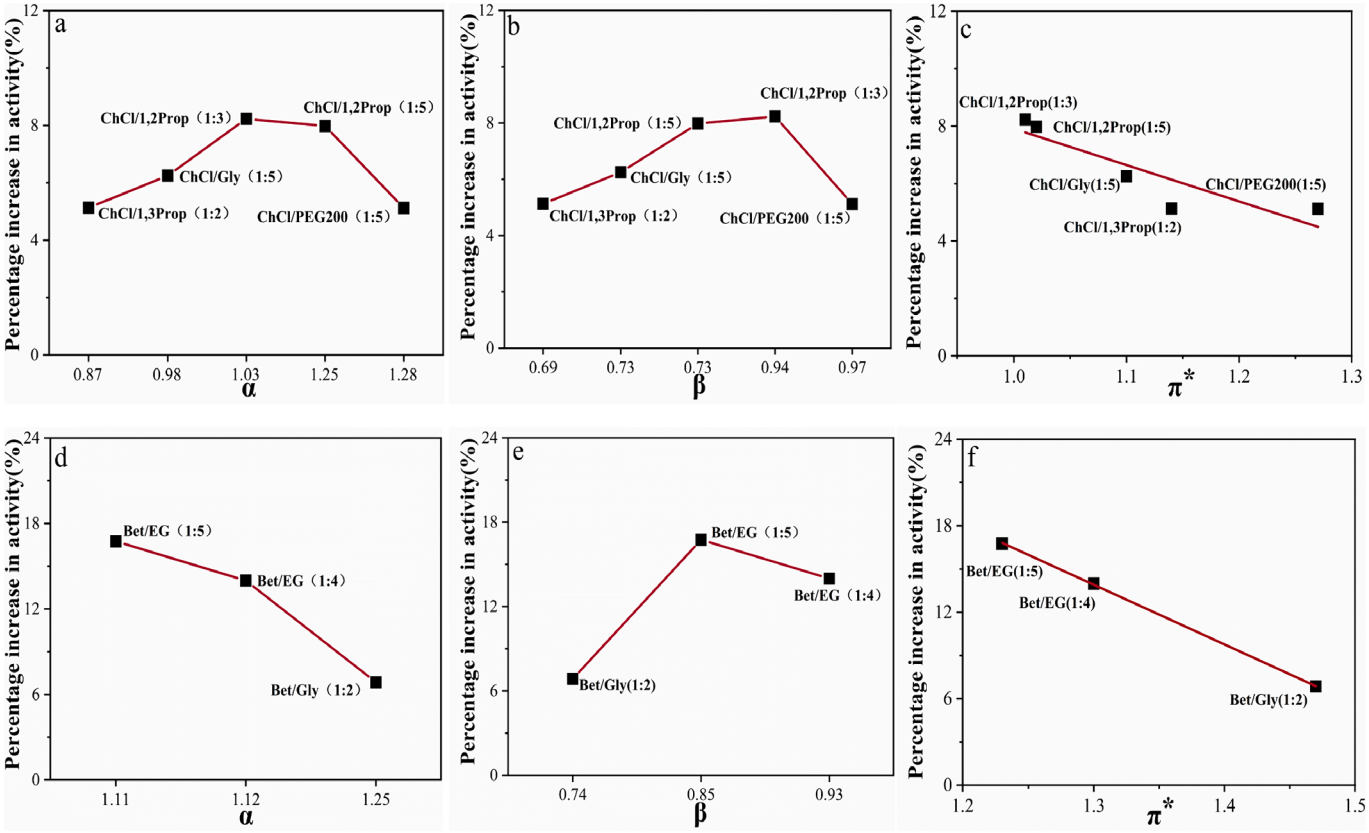


Fig. 8. The effects of Kamlet–Taft values of ChCl-DESs (a–c) and Bet-DESs (d–f) on the improvement of cellulase activity.

higher stability provide longer protection for the enzyme molecules, resulting in longer preservation of the active conformation of the enzyme.

3.3.4. *The molecular dynamics simulation of βG*

It was demonstrated that the βG activity was significantly improved in DES/AA-SA solutions, which is the main factor contributing to the enhancement of cellulase activity. In view of the significant dual enhancement effect of ChCl/PEG200 (1:5) and Bet/Gly (1:2) on the activity (Fig. 2) and stability of βG (Table S5), the molecular MD simulation was employed to investigate the structural changes of βG (PDB:3ZYZ) upon introducing these two DESs into AA-SA solution at a concentration of 10 % (v/v) and 5 % (v/v), respectively.

Fig. 10 depicts the MD simulations, illustrating the overlap of tertiary structures among different conformers of βG (a–c) and the variations in the composition of its secondary structures (d–f), including α-helix (in red), β-sheet (in yellow), and loop (in green). The tertiary structure of βG following the introduction of ChCl/PEG200 (1:5) (Fig. 10a) and Bet/Gly (1:2) (Fig. 10b) did not align well with the initial structure in the AA–SA solution. Instead, the βG structure contracted inward. This indicates that the wrapping/encapsulation by the DES increased the compactness of the overall enzyme structure. In contrast, the spatial structure of βG showed relatively better overlap after the introduction of the above two DESs (Fig. 10c), suggesting that the two DESs induced similar degrees of change in the compactness of βG.

The secondary structure of βG showed some differences in different solutions. The introduction of ChCl/PEG200 (1:5) (Fig. 10e) induced new α-helix structures in regions A1, A2, A3, and A4, while the introduction of Bet/Gly (1:2) (Fig. 10f) led to new α-helix structures in A1, A2, and A3. The additional α-helix structures observed in the DES-containing solutions primarily originated from loop structures in the tertiary structure of the enzyme in the AA–SA solution. Furthermore, the α-helix in region A5 changed differently, becoming longer in the presence of ChCl/PEG200 (1:5) (Fig. 10e) but disappearing in the presence of Bet/Gly (1:2) (Fig. 10f).

The addition of DESs resulted in a reduction in the β-sheet structure of βG. The ChCl/PEG200 (1:5) (Fig. 10e) caused the disappearance of two β-sheet structures in region B1, whereas the Bet/Gly (1:2) (Fig. 10f) resulted in the loss of one β-sheet structure in region B1. ChCl/PEG200 (1:5) (Fig. 10e) caused the disappearance of the β-sheet structure in region B2, while Bet/Gly (1:2) (Fig. 10f) increased the length of the β-sheet structure in region B2. Both ChCl/PEG200 (1:5) and Bet/Gly (1:2) transformed the fully folded β-sheet structure in region B3 into a partially folded sstructure with a shortened length.

Fig. S2 shows the changes in the secondary structure of βG induced by the two DESs. Structural analysis using SuperMDA software revealed the specific contents of the secondary structures (Table 2). The addition of the two DESs increased the α-helix content from 18.7 % (AA–SA solution) to 22.0 % (ChCl/PEG200 (1:5)) and 21.2 % (Bet/Gly (1:2)). Correspondingly, some loop structures such as coils and bends exhibited a reduction in content (Table 2). The β-sheet structure content also reduced from 21.2 % (AA–SA solution) to 18.5 % (ChCl/PEG200 (1:5)) and 19.5 % (Bet/Gly (1:2)). Correspondingly, the content of some turn structures increased (Table 2).

The radius of gyration (RG) was employed to assess the compactness of the enzyme conformation in different solutions [74] (Fig. 11a). In the AA–SA solution, the initial RG (0 ns) of βG was 2.55 nm, and the RG value stabilized at ~2.60 nm after 100 ns. Following the addition of ChCl/PEG200 (1:5) or Bet/Gly (1:2), the initial RG of βG decreased to 2.52 nm, and the RG values stabilized at 2.56 and 2.57 nm, respectively, after 100 ns. DES addition resulted in a small reduction in the average RG of the enzymes compared with the initial conformations, making the spatial structures more compact. This is attributable to the conformational contraction due to interactions formed between the DES and the enzyme molecules.

The root mean square deviation (RMSD) value can reflect the extent

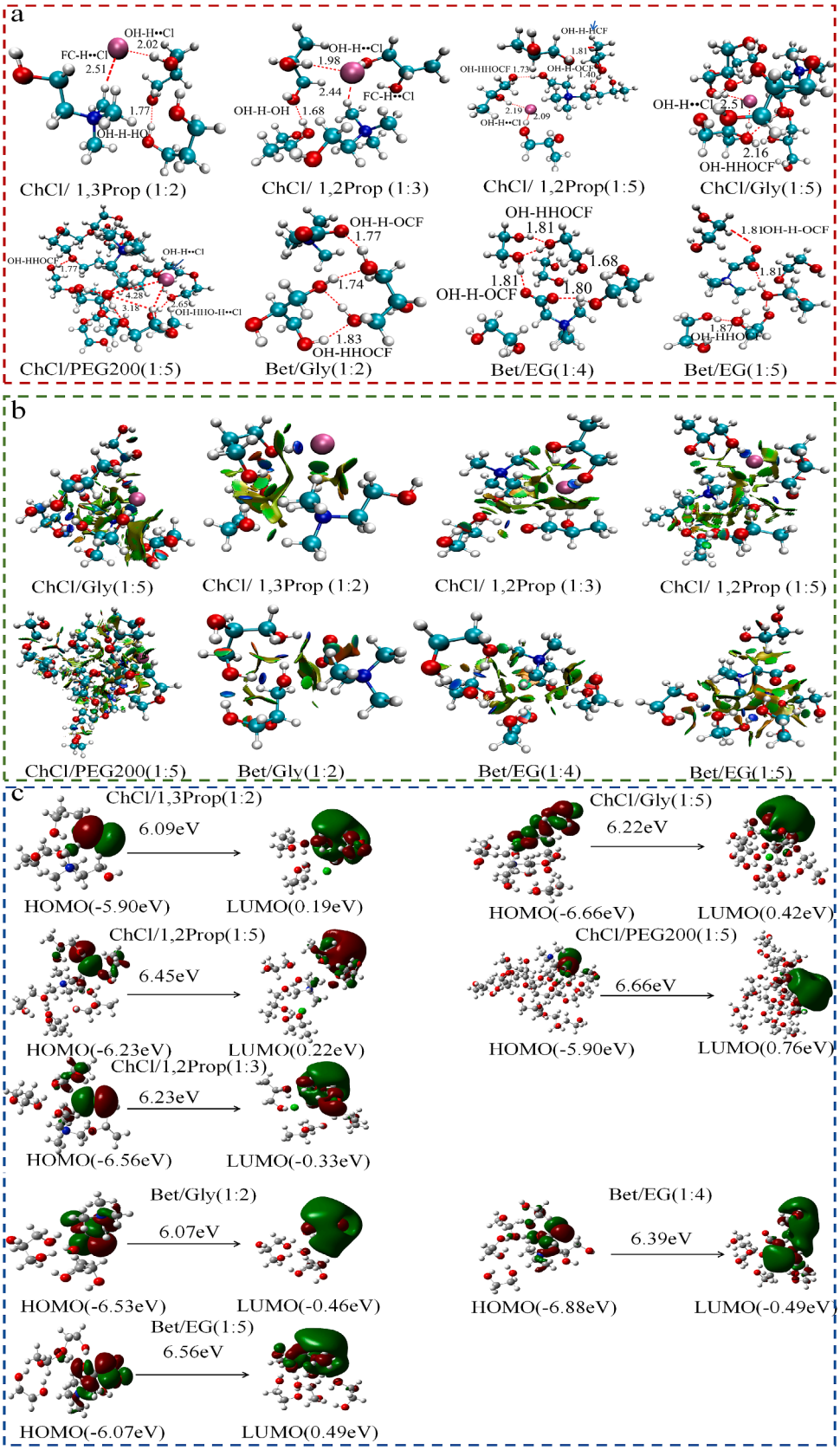


Fig. 9. The DFT calculation of DESs: a, the typical structure, hydrogen bond and bond length of DESs; b, the RDG analyses of DESs; c, the HOMO-LUMO energy level diagram of DESs (green clouds are positive, red clouds are negative); the effect of the energy gap change of ChCl-DESs (d) and Bet-DESs (e) on the half-life of cellulase.

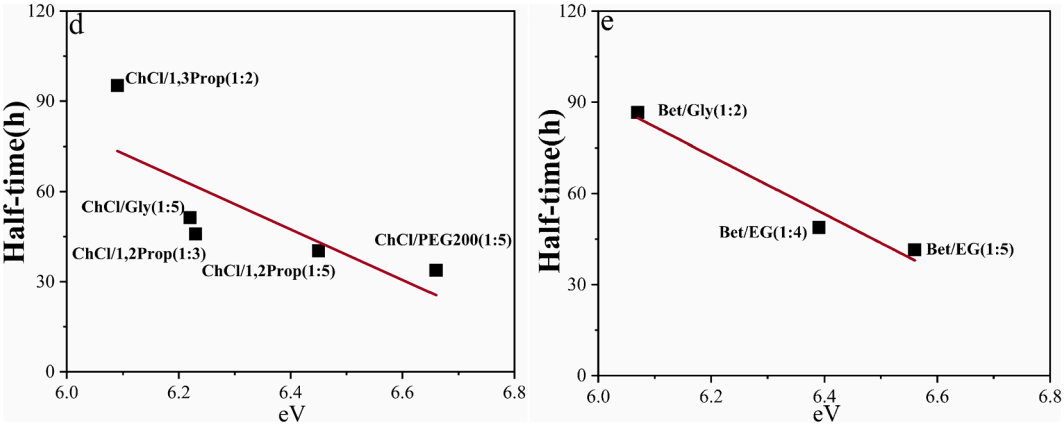


Fig. 9. (*continued*).

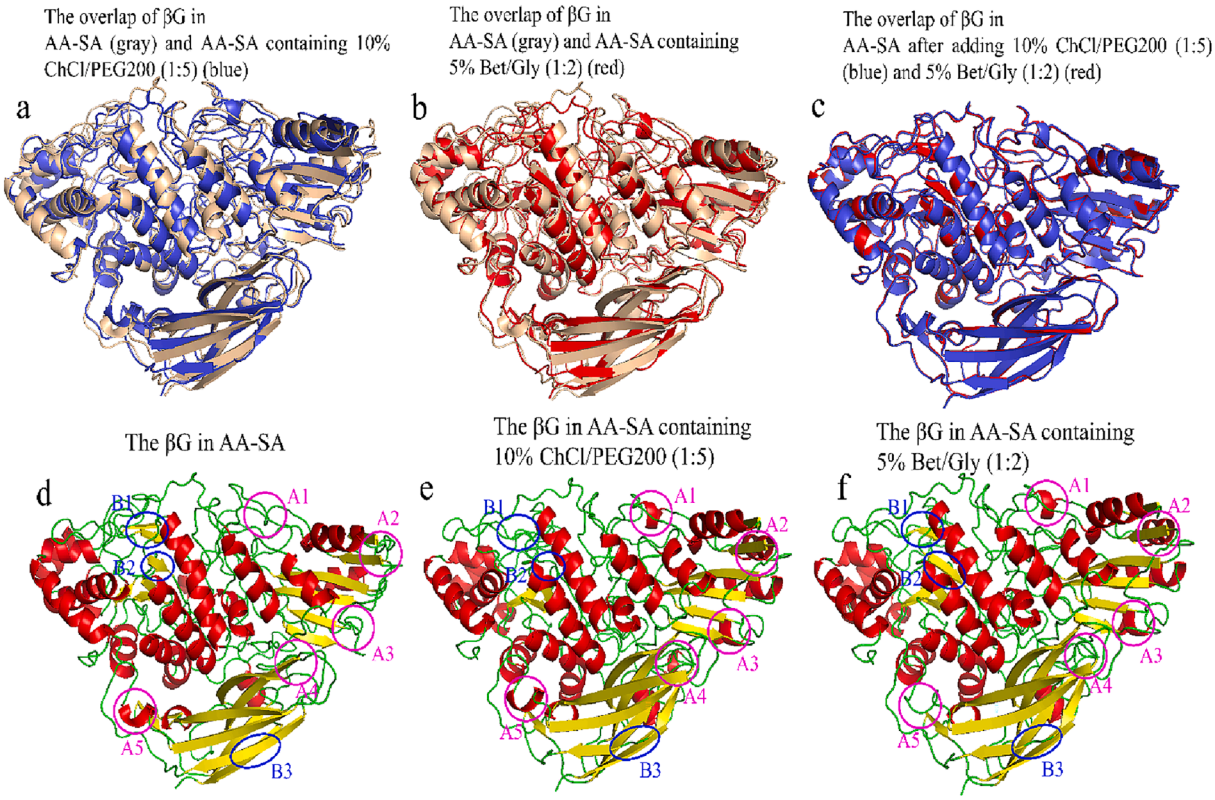


Fig. 10. The tertiary structure of βG (PDB:3ZYZ) after DES introduction, as obtained from the MD simulation: a–c, overlap between different conformers; d–f, variations in secondary structures. The amino acid residues in the A and B regions are as follows: A1: ASN360, HIS361, ALN362, and ARG363; A2: GLU489, GLY490, ILE491, LEU492, and ALA493; A3: PRO536, ASN537, and ASP538; A4: SER608, ASP609, LEU610, and PHE611; A5: THR418, SER419, SER420, GLY421, ALA422, SER423, ALA424, ALA424, GLY662, and GLN663. B1: CYS58, LEU59, GLN60, VAL108, ILE109, LEU110, and GLY111; B2: ARG586 and LEU587.

Table 2
The secondary structure contents of βG following the introduction of DESs into the AA–SA solution, as obtained from MD simulation.

System	Concentration	Coil	β-Sheet	β-Bridge	Bend	Turn	α-Helix	5-Helix	3-Helix
AA-SA	–	23.9	21.2	2.2	17.5	11.7	18.7	1.41	4.9
ChCl/PEG200 (1:5)	10 % (v/v)	22.5	18.5	2.2	15.7	12.6	22	1	5.5
Bet/Gly (1:2)	5 % (v/v)	22.2	19.5	2.1	13.7	14	21.2	1.4	5.9

to which atoms deviate from their average positions and indicates the magnitude of atomic motion [75]. A higher RMSD value indicates a larger spatial range of atomic motion and therefore less spatial constraint [76]. Fig. 11b describes the RMSD values of the Cα atoms (backbone carbon atoms) of βG in different solutions. The RMSD values of the Cα atoms gradually converged to a constant value over the simulation time, fluctuating around a certain value, indicating that the system had reached energy equilibrium. After 100 ns, the RMSD value

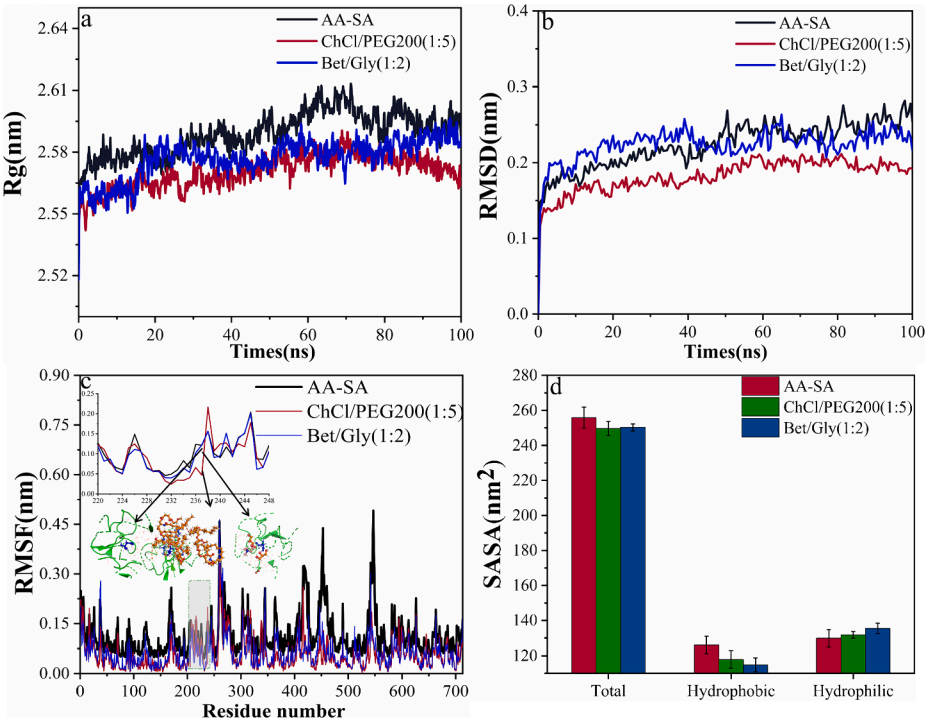


Fig. 11. The Rg (a), RMSD (b), RMSF (c) and SASA (d) values of β-glucosidase in DES/AA-SA solution over the entire simulation time of 100 ns.

Table 3
The average number of hydrogen bonds of β-glucosidase after DES addition to AA–SA solution, as obtained via MD simulation.

System	Concentration	Protein-Protein	Protein-HBA	Protein-HBD	Protein-water
AA-SA	–	512.4	–	–	1149.4
ChCl/PEG200 (1:5)	10 % (v/v)	537.9	3	27	1029.4
Bet/Gly (1:2)	5 % (v/v)	521.8	2.6	16	1117.4

was ~ 0.26 nm for βG in the AA–SA solution, 0.24 nm after the introduction of Bet/Gly (1:2), and 0.21 nm after the introduction of ChCl/PEG200 (1:5). The introduction of the DES reduced the RMSD value of the Cα atom, indicating that the sheathing effect of the DES reduced the motion range of the backbone carbon atoms and improved the stability of the chemical structure. Furthermore, the lower RMSD value showed that ChCl-PEG200 imparted greater stability than Bet-Gly (1:2).

The root mean square fluctuation (RMSF) value is an important indicator of the mobility of amino acid residues in proteins [77]. In the βG protein, most of the amino acid residues exhibited an RMSF value below 0.25 nm in the AA–SA solution (Fig. 11c), while specific regions (positions 10–15, 22–45, 160–180, and 260–270) exhibited higher RMSF values, indicating relatively higher mobility in these areas. Upon the addition of ChCl/PEG200 (1:5) and Bet/Gly (1:2), the RMSF values of most amino acid residues decreased, except for positions 43–50 and 260–270, where the RMSF values increased. This phenomenon also reflects the effect of DESs on the structure of the βG protein. The DESs compacted the protein structure spatially. Notably, the introduction of the two DESs also reduced the RMSF value at the active site, specifically at the aspartic acid (ASP) residue (236), in βG. This suggests that DES immobilized the ASP residue through hydrogen bonding (Fig. 11c inset), thereby reducing its flexibility. Consequently, the active site conformation of βG was stabilized, enhancing the overall structural stability and catalytic activity of the protein [78].

The hydrogen bonding interactions between proteins and solvents play a vital role in stabilizing enzyme conformations [74,79]. Table 3 presents the types and average number of hydrogen bonds of βG in different solutions. The AA–SA solution exhibited two main types of bonds: intramolecular protein hydrogen bonds and protein–water hydrogen bonds. With the introduction of the two DESs, the protein-–water hydrogen bonds decreased, suggesting that DES molecules replaced some water molecules to form hydrogen bonds with proteins, effectively encapsulating the enzymes. This wrapping effect enhanced intramolecular hydrogen bonding within the enzymes, leading to an increase in the number of such bonds. Notably, the number of hydrogen bonds formed between the HBD end of the DES and proteins was significantly higher than that formed between the HBA end and proteins. This discrepancy is attributable to the hydroxyl groups in the HBD end, which readily form hydrogen bonds with proteins [75].

The protein surface area can indicate alterations in the folding degree of protein structures and the relative distribution of hydrophobic and hydrophilic amino acid residues [60]. Fig. 11d illustrates the solvent-accessible surface area (SASA) of βG and its hydrophobic and hydrophilic surface areas. In the AA–SA solution, βG exhibited a total surface area of 254.54 nm^2. After the introduction of the two DESs, the total area βG decreased to 249.59 nm^2 (ChCl/PEG200 (1:5)) and 250.24 nm^2 (Bet/Gly (1:2)). The decrease in surface area further confirms the compactness of βG. Additionally, the two DESs significantly reduced the hydrophobic surface area of βG. Fluorescence analysis indicated that the DESs caused the movement of hydrophobic Trp residues toward the interior of the enzyme molecule, which may be the crucial factor contributing to the reduction in hydrophobic surface area. Notably, the introduction of the two DESs increased the hydrophilic surface area of βG. ChCl/PEG200 (1:5) resulted in a 2 nm^2 increase, while Bet-DESd resulted in a 5.6 nm^2 increase in the hydrophilic surface area. The increase in hydrophilicity facilitated the binding ability of enzymes to substrates, thereby enhancing the activity of the enzymes.

3.4. Enzymatic hydrolysis of lignocellulosic biomass

The novel DES/AA-SA solutions were employed in the hydrolysis of

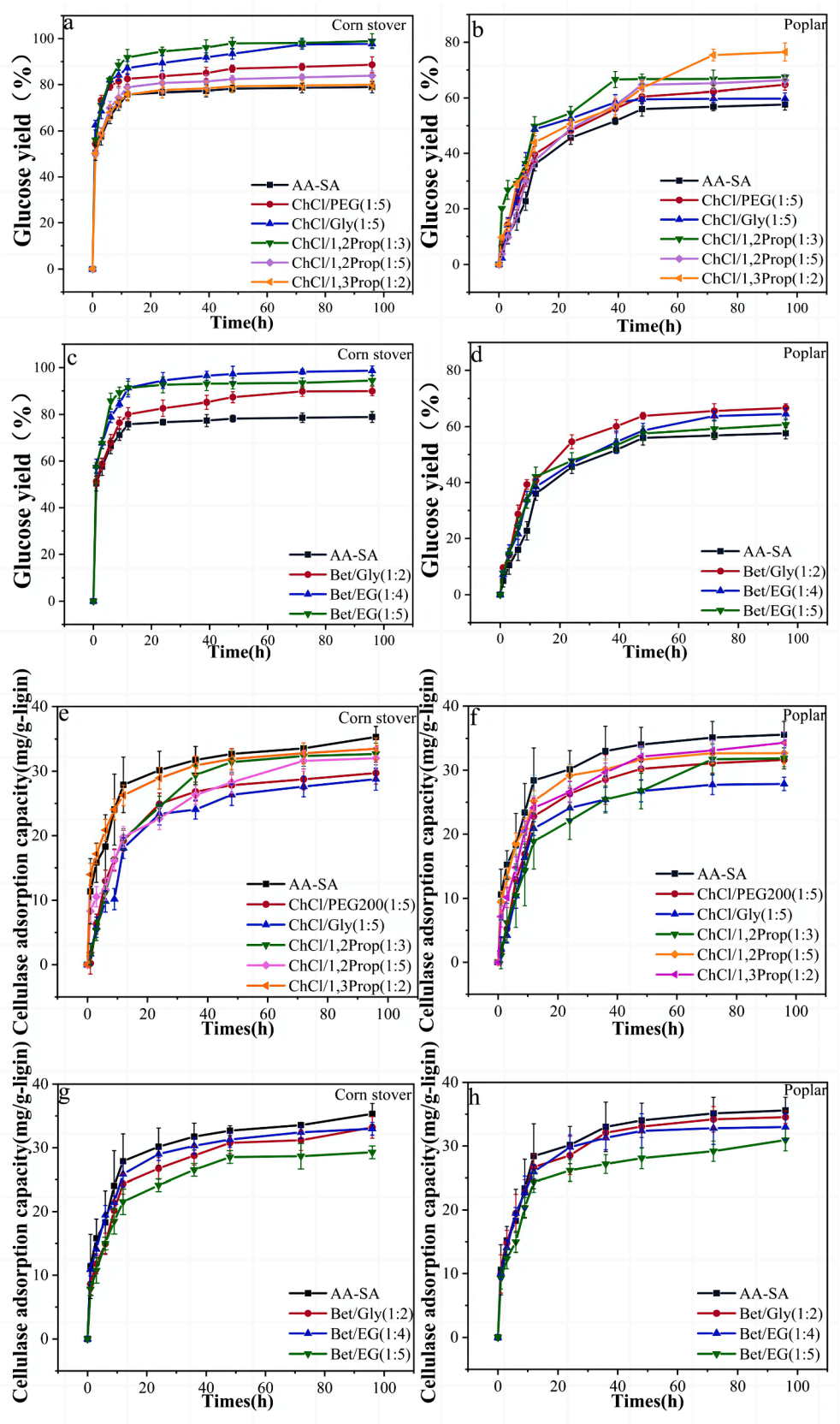


Fig. 12. The glucose yield during enzymatic hydrolysis of corn stover and poplar in ChCl-DES/AA-SA solutions (a and b) and Bet-DES/AA-SA solutions (c and d), and the unproductive adsorption capacity of lignin on cellulase in ChCl-DES/AA-SA solutions (e and f) and Bet-DES/AA-SA solutions (g and h).

lignocellulosic biomass, such as corn stover and poplar, to generate glucose (Fig. 12a–d). In addition to enzyme activity, the unproductive adsorption of enzymes on lignin also influenced the hydrolysis efficiency of cellulase (Fig. 12e–h). The corn stover and poplar wood were pretreated using a green ChCl/EG (1:5) solvent to remove lignin and hemicellulose. Under the pretreatment condition of a solid-to-liquid ratio of 1:30 at 200 °C for 3 h, the substrate exhibited lower lignin and xylan contents, while the glucan content was the highest (Tables S6 and S7).

The pretreated corn stover still contained 6.7 % acid-insoluble lignin and 0.6 % acid-soluble lignin, while the pretreated poplar wood still contained 10.2 % acid-insoluble lignin and 0.5 % acid-soluble lignin. The introduction of ChCl-DESs and Bet-DESs improved the efficiency of enzymatic saccharification for both corn stover and poplar wood to varying degrees (Fig. 12a–d). The glucose yield from pretreated corn stover in the conventional AA–SA solution was only 78 %. Owing to the higher lignin content of the pretreated poplar, its glucose yield was lower, at only 55 %. The introduction of ChCl/1,2-prop (1:3) and Bet/EG (1:4) resulted in the most significant enhancement of enzymatic hydrolysis for corn stover, with both achieving a glucose yield of 98 %. The introduction of ChCl/1,3-prop (1:2) and Bet/Gly (1:2) resulted in the most significant improvement in the enzymatic hydrolysis of poplar wood, increasing the glucose yield to 80 % and 70 % respectively. Additionally, the introduction of these DESs reduced the unproductive adsorption of enzymes on lignin (Fig. 12e–h).

Interestingly, the improvement in activity, half-life, and nonproductive adsorption capacity of lignin for enzymes when DESs were introduced was not strongly correlated with glucose yield. For example, the introduction of ChCl-DESs did not significantly improve cellulase activity, with improvements ranging from 5.1 % to 8.2 %. However, the effect of the DESs on the enzymatic hydrolysis efficiency of corn stover varied widely (Fig. 12a). The introduction of ChCl/1,3-prop (1:2), which resulted in the most significant improvement in cellulase half-life, hardly improved the enzymatic hydrolysis of corn stover. Similarly, Bet/Gly (1:2), which also resulted in the most significant improvement in cellulase half-life, did not result in optimal improvement in the enzymatic hydrolysis of corn stover. In another example, ChCl/1,3-prop (1:2) and Bet/Gly (1:2) caused the highest improvements in enzymatic hydrolysis of poplar wood, although the unproductive adsorption of enzymes on lignin was not minimized. This could be attributed to the release of lignin, especially acid soluble lignin, from the fiber interior into the solvent during the enzymatic hydrolysis process. The released lignin molecules could interact with the DES through hydrogen bonding, disrupting the protective effect of the DES on the enzymes and affecting the activity and half-life of the enzymes and the unproductive adsorption of lignin on the enzymes. This in turn would affect the efficiency of enzymatic saccharification. According to the data, corn stover lignin appeared to have a greater effect on ChCl/1,3-prop (1:2) and Bet/Gly (1:2), while poplar lignin had a lesser effect.

4. Conclusions

The incorporation of specific ChCl-DES and Bet-DES into AA–SA solution increased cellulase activity by 5 %–10 % and 6 %–13 % respectively. This increase in cellulase activity was mainly due to the increased βG activity, which was improved by 20 %–35 %. In addition, the half-life of cellulase was significantly prolonged, reaching three times the initial value. The hydrogen bonding reaction between DES and the enzyme promoted proper enzyme folding, resulting in a more compact enzyme structure by increasing the α-helix content and reducing the β-sheet content. This change resulted in reduced structural flexibility and mobility, thereby improving cellulase stability. The MD simulation on βG showed that DES interacted with the enzyme mainly at the HBD end, resulting in more compact structures, stabilizing the active site and increasing the hydrophilic surface area, which enhanced βG activity. In addition, the DESs formed a protective layer that prevented the unproductive adsorption of lignin on enzymes during the enzymatic hydrolysis of lignocellulosic biomass. These multiple enhancements resulted in significantly improved enzymatic saccharification efficiency of both corn stover and poplar.

CRediT authorship contribution statement

Yanming Wang: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Ailin Wang:** Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Shiwei Liu:** Supervision, Methodology. **Lu Li:** Supervision, Conceptualization. **Yue Liu:** Resources, Conceptualization. **Yuxiang Liu:** Software, Resources, Conceptualization. **Genkuo Nie:** Writing – review & editing, Supervision, Conceptualization. **Shuangxi Nie:** Supervision, Funding acquisition. **Shuangquan Yao:** Investigation, Funding acquisition. **Hailong Yu:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (32171735) and the Open Project of Guangxi Key Laboratory of Clean Pulp & Papermaking and Pollution Control (2021KF04).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2024.151439.

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